

LumenTrace: A high-performance MCU-based LED racing simulator

Technical Reports: CL-2026-42, July 2026

Simon Völkl, Erik Dombrowsky, Simon Frey 
Department of Electrical Engineering, Media, and Computer Science
Ostbayerische Technische Hochschule Amberg-Weiden
Amberg, Germany

Abstract—Background: tbd
Methods: tbd
Conclusion: tbd
Index Terms—MCU, LED, neopixel, WS2812, racing simulator, Raspberry Pi

1	{}
Listing 1. example listing	

I. INTRODUCTION

- A. Motivation
tbd
tbd
- B. Unique Selling Point
tbd
- C. Key Technical Components
tbd

II. RELATED WORK

tbd

III. REQUIREMENTS

- A. Project Personas
To ensure the system meets the diverse needs of all stakeholders, the requirements are guided by the following personas. Each persona has a specific role, defined goals, and distinct interests in the system:
AAA AAA tbd
- B. User Stories
 - 1) Real-time Temperature Monitoring:
yada yadaAcceptance Criteria:
 - yada yada
- C. Definition of Done
tbd
 - yada yada
- D. Prioritisation (MoSCoW)
yada yada
- E. Minimum Viable Product (MVP)
tbd

IV. TECHNICAL CONCEPT

- A. Data Acquisition and Transmission
- B. Technology Stack
 - yada yada
- C. Testability
- D. Costs

V. CONCLUSION AND FUTURE WORK